

Question 1

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(a) Find the value of $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ in terms of π . 1

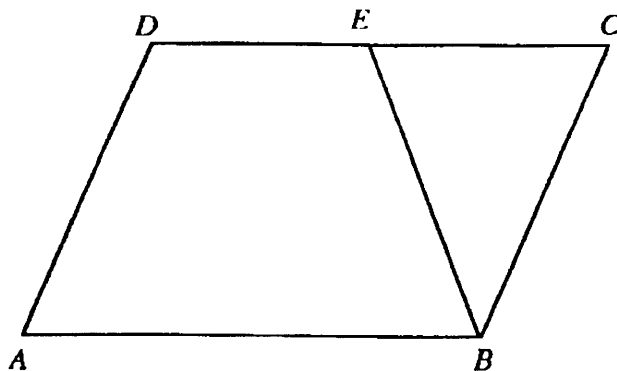
(b) The acute angle between the line $x - 2y + 3 = 0$ and the line $y = mx$ is 45° . 3

(i) Show that $\left|\frac{2m-1}{m+2}\right| = 1$

(ii) Find the possible values of m .

(c) Solve the equation $\ln(x^2 + 19) = 2 \ln(x + 1)$. 3

(d) 5



$ABCD$ is a parallelogram. E is the point on CD such that $BE = BC$.

(i) Copy the diagram showing the above information.

(ii) Show that $ABED$ is a cyclic quadrilateral.

Question 2

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- (a) Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$ 1
- (b) Solve the inequality $\frac{x^2 + 9}{x} \leq 6$ 3
- (c) (i) Factorise $3x^3 + 3x^2 - x - 1$ 3
- (ii) Solve the equation $3 \tan^3 \theta + 3 \tan^2 \theta - \tan \theta - 1 = 0$ for $0 \leq \theta \leq \pi$
- (d) $P(2t, t^2)$ is a point on the parabola $x^2 = 4y$ with focus F . The point M divides the interval FP externally in the ratio 3 : 1. 5
- (i) Show that as P moves on the parabola $x^2 = 4y$, then M moves on the parabola $x^2 = 6y + 3$.
- (ii) Find the coordinates of the focus and the equation of the directrix of the locus of M .

Question 3

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- (a) Find the gradient of the tangent to the curve $y = \tan^{-1} \frac{1}{x}$ at the point on the curve where $x = 1$. 2
- (b) A function is given by the rule $f(x) = \frac{x+1}{x+2}$. Find the rule for the inverse function $f^{-1}(x)$. 2
- (c) At any point on the curve $y = f(x)$ the gradient function is given by $\frac{dy}{dx} = 2\cos^2 x + 1$. 4
If $y = \pi$ when $x = \pi$, find the value of y when $x = 2\pi$.
- (d) Use the substitution $x = u^2$, $u > 0$, to express the value of $\int_1^{100} \frac{1}{x+2\sqrt{x}} dx$ 4
in the form $\ln a$ for some constant $a > 0$.

Question 4**Begin a new page**

- (a) Find the exact value of $\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$. 2
- (b) A particle is moving in a straight line. At time t seconds its displacement x metres from a fixed point O on the line is such that $t = x^2 - 3x + 2$. 2
- (i) Find an expression for its velocity v in terms of x .
- (ii) Find an expression for its acceleration a in terms of x .
- (c) Consider the function $y = 2\cos^{-1}(1-x)$. 4
- (i) Find the domain and range of the function.
- (ii) Sketch the graph of the function.
- (d) The radius r kilometres of a circular oil spill at time t hours after it was first observed is given by $r = \frac{1+3t}{1+t}$. Find the exact rate of increase of the area of the oil spill when the radius is 2 kilometres. 4

Question 5**Begin a new page**

- (a) Consider the function $f(x) = \frac{\ln x}{x}$. 6
- (i) Find the coordinates and the nature of the stationary point on the curve $y = f(x)$.
- (ii) Explain why $f(\pi) < f(e)$ and hence show that $\pi^e < e^\pi$.
- (iii) $P(X, -2)$ is a point on the curve $y = f(x)$. Starting with an initial approximation of $X = 0.5$, use one application of Newton's method to find an improved approximation to the value of X , giving the answer correct to 2 decimal places.

Question 5 (Cont)

- (b) A machine which initially costs \$49 000 loses value at a rate proportional to the difference between its current value $\$M$ and its final scrap value $\$1000$. After 2 years the value of the machine is $\$25\,000$. 6
- (i) Explain why $\frac{dM}{dt} = -k(M - 1000)$ for some constant $k > 0$, and verify that $M = 1000 + Ae^{-kt}$, A constant, is a solution of this equation.
- (ii) Find the exact values of A and k .
- (iii) Find the value of the machine, and the time that has elapsed, when the machine is losing value at a rate equal to one quarter of the initial rate at which it loses value.

Question 6

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- (a) If α , β and χ are the roots of $3x^3 + 5x^2 - 7x + 4 = 0$, find the values of 2
- (i) $\alpha + \beta + \chi$
- (ii) $\alpha\beta + \alpha\chi + \beta\chi$
- (b) Two circles touch internally at a point P. A line through P cuts the smaller circle at A and the larger circle at B. A second line through P cuts the smaller and larger circles at C and D respectively. 4
- (i) Draw a diagram showing this information.
- (ii) Prove that AC is parallel to BD.
- (c) A particle moving in a straight line is performing Simple Harmonic Motion. At time t seconds its displacement x metres from a fixed point O on the line is given by $x = 2\sin 3t - 2\sqrt{3}\cos 3t$. 6
- (i) Express x in the form $x = R\sin(3t - \alpha)$ for some constants $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.
- (ii) Describe the initial motion of the particle in terms of its initial position, velocity and acceleration
- (iii) Find the exact value of the first time that the particle is 2 metres to the left of O and moving towards O .

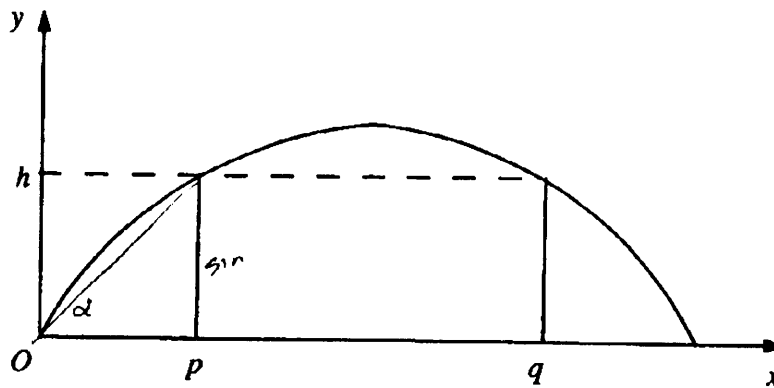
Question 7

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(a) Use the method of mathematical induction to prove that $7^n - 5^n$ is even, for all positive integers $n \geq 1$. 4

(b) Given that ABCD is a cyclic quadrilateral, show that
 $\tan A + \tan B + \tan C + \tan D = 0$ 2

(c) 6



A particle is projected with velocity $V \text{ ms}^{-1}$ from a point O at an angle of elevation α . Axes Ox and Oy are taken horizontally and vertically through O . The particle just clears two vertical chimneys of height h metres at horizontal distances of p metres and q metres from O . The acceleration due to gravity is taken as 10 ms^{-2} and air resistance is ignored.

(i) Write down expressions for the horizontal displacement x and the vertical displacement y of the particle after time t seconds.

(ii) Show that $V^2 = \frac{5p^2(1 + \tan^2 \alpha)}{p \tan \alpha - h}$.

(iii) Show that $\tan \alpha = \frac{h(p+q)}{pq}$.

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① (a) $\frac{5\pi}{6}$

(b) $x - 2y + 3 = 0$ has gradient $\frac{1}{2}$

$\therefore \tan 45^\circ = \left| \frac{m - \frac{1}{2}}{1 + \frac{m}{2}} \right|$

$1 = \left| \frac{2m - 1}{2 + m} \right|$

(i) $\frac{2m - 1}{m + 2} = 1$ or $\frac{2m - 1}{m + 2} = -1$

$2m - 1 = m + 2$ $2m - 1 = -m - 2$
 $m = 3$ $3m = -1$

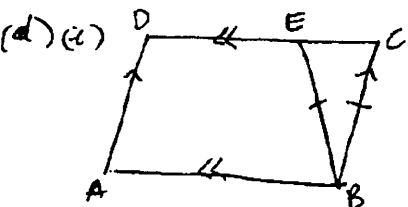
$\therefore m = 3$ or $-\frac{1}{3}$

(c) $\ln(x^2 + 19) = \ln(x + 1)^2$

$x^2 + 19 = x^2 + 2x + 1$

$18 = 2x$

$x = 9$



(ii) $\angle BCE = \angle BEC$ (equal angles opposite equal sides in $\triangle BCE$)

Also $\angle BCE = \angle BAD$ (opposite angles of parallelogram)

$\therefore \angle BEC = \angle BAD$

$\therefore ABED$ is a cyclic quadrilateral

(exterior angle equal to opposite interior angle)

② (a) $\lim_{x \rightarrow 0} \frac{\sin 2x}{x} = 2 \lim_{x \rightarrow 0} \frac{\sin x}{2x}$

$= 2 \times 1$

$= 2$

(b) $\frac{x^2 + 9}{x} \times x^2 \leq 6x^2$

$x^3 + 9x \leq 6x^2$

$x^3 - 6x^2 + 9x \leq 0$

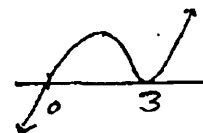
$x(x^2 - 6x + 9) \leq 0$

$x(x - 3)^2 \leq 0$

$x \leq 0$ or $x = 3$

But $x \neq 0$

$\therefore x < 0$ or $x = 3$



(c) $3x^2(x+1) - 1(x+1) = (x+1)(3x^2 - 1)$

(i) Let $x = \tan \theta$

$\therefore (\tan \theta + 1)(3 \tan^2 \theta - 1) = 0$

$\tan \theta = -1$ or $\tan^2 \theta = \frac{1}{3}$, $0 \leq \theta < \pi$

$\tan \theta = \pm \frac{1}{\sqrt{3}}$

$\theta = \frac{3\pi}{4}, \frac{\pi}{6}, \frac{5\pi}{6}$

(d) (i) $x = \frac{3 \times 2t - 1 \times 0}{3 - 1}$, $y = \frac{3 \times t^2 - 1 \times 1}{3 - 1}$

$= \frac{6t}{2}$

$x = 3t$ (1)

$y = \frac{3t^2 - 1}{2}$ (2)

From (1), $t = \frac{x}{3}$, so from (2), $y = \frac{3 \cdot \frac{x^2}{9} - 1}{2}$

$2y = \frac{x^2}{3} - 1$

$x^2 = 6y + 3$

(ii) $x^2 = 6(y + \frac{1}{2})$ Focus $(0, 1)$, dir $y = -2$

$$3(a) y = \tan^{-1} x$$

$$\frac{dy}{dx} = \frac{1}{1+x^2} \cdot -\frac{1}{x^2}$$

$$= \frac{-1}{x^2+1}$$

When $x=1$, $\frac{dy}{dx} = -\frac{1}{2}$ = gradient of tangent

(b) Inverse is $x = \frac{y+1}{y+2}$

$$xy+2x = y+1$$

$$y(x-1) = 1-2x$$

$$\therefore y = \frac{1-2x}{x-1}$$

$$\therefore f^{-1}(x) = \frac{1-2x}{x-1}$$

(c) $\frac{dy}{dx} = 2\cos^2 x + 1$

$$= 1 + \cos 2x + 1$$

$$\cos 2x = 2\cos^2 x - 1$$

$$= 2 + \cos 2x$$

$$y = 2x + \frac{1}{2} \sin 2x + c$$

When $x=\pi$, $y=\pi$

$$\therefore \pi = 2\pi + \frac{1}{2} \sin 2\pi + c$$

$$\therefore c = -\pi$$

$$y = 2x + \frac{1}{2} \sin 2x - \pi$$

When $x=2\pi$, $y = 4\pi + \frac{1}{2} \sin 4\pi - \pi$

$$= 3\pi$$

(d) $x = u^2$

$$\frac{dx}{du} = 2u$$

$$dx = 2u \cdot du$$

When $x=1$, $u=1$

" $x=100$, $u=10$

$$\therefore \int_1^{10} \frac{2u \cdot du}{u^2+2u} = \int_1^{10} \frac{2 du}{u+2}$$

$$= 2 [\ln(u+2)]_1^{10} = 2 \ln \frac{12}{3}$$

$$= \ln 16$$

(4) (a) $\int_{\frac{1}{2}\sqrt{4-x^2}}^{\frac{1}{2}} \frac{dx}{\sqrt{4-x^2}} = \left[\sin^{-1} \frac{x}{2} \right]_{\frac{1}{2}}^{\frac{1}{2}}$

$$= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{\sqrt{2}}{2}$$

$$= \frac{\pi}{3} - \frac{\pi}{4}$$

$$= \frac{\pi}{12}$$

(b) (i) $t = x^2 - 3x + 2$

$$\frac{dt}{dx} = 2x - 3$$

$$\therefore v = \frac{dx}{dt} = \frac{1}{2x-3}$$

(ii) $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$$= \frac{d}{dx} \left[\frac{1}{2} (2x-3)^{-2} \right]$$

$$= -(2x-3)^{-3} \cdot 2$$

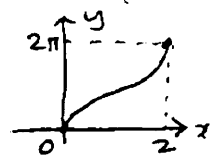
$$= -\frac{2}{(2x-3)^3}$$

(c) (i) Domain is $-1 \leq 1-x \leq 1$

$$-2 \leq -x \leq 0$$

$$0 \leq x \leq 2$$

Range is $0 \leq y \leq 2\pi$



(d) $\frac{dr}{dt} = \frac{(1+t) \cdot 3 - (1+3t) \cdot 1}{(1+t)^2}$ and $A = \pi r^2$

$$= \frac{2}{(1+t)^2}$$

$$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$$

$$= \frac{4\pi r}{(1+t)^2}$$

When $r=2$, $2 = \frac{1+3t}{1+t}$

$$2+2t = 1+3t$$

$$t=1$$

$$\therefore \frac{dA}{dt} = \frac{8\pi}{4} = 2\pi$$

$\therefore$ Rate of increase is $2\pi \text{ km/h}$

$$\begin{aligned}
 5(a) (i) \frac{dy}{dx} &= \frac{x \cdot \frac{1}{x} - \ln x \cdot 1}{x^2} \\
 &= \frac{1 - \ln x}{x^2} \\
 \frac{d^2y}{dx^2} &= \frac{x^2 \cdot -\frac{1}{x^3} - (1 - \ln x) \cdot 2x}{x^4} \\
 &= \frac{2x \ln x - 3x}{x^4} \\
 &= \frac{2 \ln x - 3}{x^3}
 \end{aligned}$$

$$\begin{aligned}
 \frac{dy}{dx} = 0 &\Rightarrow \frac{1 - \ln x}{x^2} = 0 \\
 \ln x &= 1 \\
 x &= e
 \end{aligned}$$

$$\text{When } x = e, \frac{d^2y}{dx^2} = -\frac{1}{e^3} < 0$$

$\therefore$ Max. turning pt at $(e, \frac{1}{e})$

(ii) Since there is a maximum at $x = e$, and $\frac{dy}{dx} < 0$ for $x > e$, then, since $\pi > e$,

$$\begin{aligned}
 f(\pi) &< f(e) \\
 \text{i.e. } \frac{\ln \pi}{\pi} &< \frac{\ln e}{e} \\
 e \ln \pi &< \pi \ln e \\
 \ln \pi^e &< \ln e^\pi \\
 \text{i.e. } \pi^e &< e^\pi
 \end{aligned}$$

$$\begin{aligned}
 (iii) \frac{\ln x}{x} &= -2 \\
 \ln x &= -2x \\
 \ln x + 2x &= 0 \\
 \text{Let } p(x) &= \ln x + 2x \\
 p'(x) &= \frac{1}{x} + 2
 \end{aligned}$$

$$\begin{aligned}
 \text{If } x_1 &= 0.5, \\
 x_2 &= 0.5 - \frac{p(0.5)}{p'(0.5)} \\
 &= 0.42 (2dp)
 \end{aligned}$$

$$(b) (i) \frac{dM}{dt} \propto M - 1000 \text{ and } \frac{dM}{dt} < 0$$

$$\therefore \frac{dM}{dt} = -k(M - 1000) \quad (k > 0)$$

$$M = 1000 + Ae^{-kt}$$

$$\begin{aligned}
 \frac{dM}{dt} &= -Ake^{-kt} \\
 &= -k(M - 1000)
 \end{aligned}$$

$$(ii) \text{ When } t = 0, M = 49000$$

$$\therefore 49000 = 1000 + A$$

$$\therefore A = 48000$$

$$\text{Then } M = 1000 + 48000e^{-kt}$$

$$\text{When } t = 2, M = 25000$$

$$25000 = 1000 + 48000e^{-2k}$$

$$24000 = 48000e^{-2k}$$

$$0.5 = e^{-2k}$$

$$-2k = \ln 0.5$$

$$-k = \frac{-\ln 0.5}{2}$$

$$= \frac{\ln 2}{2}$$

$$(iii) \text{ When } M = 49000,$$

$$\frac{dM}{dt} = -\frac{\ln 2}{2}(49000 - 1000)$$

$$= -24000 \ln 2$$

$$\text{Let } \frac{dM}{dt} = \frac{-24000 \ln 2}{4} = \frac{-\ln 2}{2}(M - 1000)$$

$$12000 = M - 1000$$

$$\therefore M = 13000$$

$$\text{When } M = 13000, 13000 = 1000 + 48000e^{-kt}$$

$$12000 = 48000e^{-kt}$$

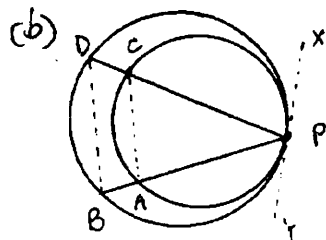
$$0.25 = e^{-kt}$$

$$-kt = \ln 0.25$$

$$t = \frac{-\ln 0.25}{k} = 4$$

$$(i) (a) \alpha) 2 + \beta + \gamma = -\frac{5}{3}$$

$$(ii) 2\beta + 2\gamma + \beta\gamma = -\frac{7}{3}$$



(ii) Draw common tangent through P.
Call it XY.

Then $\angle XPC = \angle PAC$ (angle between tangent and chord equal to angle in alternate segment)

and $\angle XPC = \angle PBD$ for large circle

$$\therefore \angle PAC = \angle PBD$$

$\therefore AC \parallel BD$ (corresponding angles equal)

$$(c) (i) \text{ Let } 2 \sin 3t - 2\sqrt{3} \cos 3t = R \sin(3t - \alpha) \\ = R(\sin 3t \cos \alpha - \cos 3t \sin \alpha)$$

$$\therefore \begin{cases} R \cos \alpha = 2 \\ R \sin \alpha = 2\sqrt{3} \end{cases}$$

$$R^2 = 2^2 + (2\sqrt{3})^2 \text{ and } \tan \alpha = \sqrt{3} \\ = 4 + 12$$

$$\therefore R = 4 \text{ and } \alpha = \frac{\pi}{3}$$

$$\therefore x = 4 \sin(3t - \frac{\pi}{3})$$

$$(ii) \dot{x} = 12 \cos(3t - \frac{\pi}{3}) \\ \ddot{x} = -36 \sin(3t - \frac{\pi}{3})$$

$$\text{When } t=0, x = -2\sqrt{3} \\ \dot{x} = 6 \\ \ddot{x} = 18\sqrt{3}$$

$\therefore$ Initially, the particle is 2.3 m to the left of O, moving at 6 m/s to the right, speeding up at a rate of $18\sqrt{3} \text{ m/s}^2$.

$$(iii) \text{ When } x = -2, -2 = 4 \sin(3t - \frac{\pi}{3})$$

$$\sin(3t - \frac{\pi}{3}) = -\frac{1}{2}$$

$$3t - \frac{\pi}{3} = -\frac{\pi}{6}, \frac{7\pi}{6}, \dots$$

$$3t = \frac{\pi}{6}, \frac{3\pi}{2}, \dots$$

$$t = \frac{\pi}{18}, \frac{\pi}{2}, \dots$$

$$\text{When } t = \frac{\pi}{18}, \dot{x} = 12 \cos(\frac{\pi}{6} - \frac{\pi}{3}) \\ = 6\sqrt{3} > 0$$

$\therefore$ First time is $\frac{\pi}{18}$ seconds.

$$(7) (a) \text{ When } n=1, 7^1 - 5^1 = 2, \text{ which is even}$$

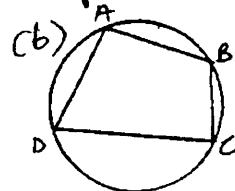
$\therefore$ True for $n=1$

Assume true for $n=k$, i.e. $7^k - 5^k = 2p$, where p is a positive integer

$$\text{When } n=k+1, 7^{k+1} - 5^{k+1} = 7 \cdot 7^k - 5 \cdot 5^k \\ = 7(2p + 5^k) - 5 \cdot 5^k \text{ using the assumption} \\ = 14p + 2 \cdot 5^k \\ = 2(7p + 5^k)$$

which is divisible by 2, as $7p + 5^k$ is a pos. int.

$\therefore$ True for $n=k+1$ if true for $n=k$. Since true for $n=1$, then true for all integers $n \geq 1$.



$$C = 180^\circ - A \text{ and } D = 180^\circ - B \text{ (opposite angles of cyclic quadrilateral supplementary)} \\ \therefore \tan A + \tan B + \tan C + \tan D \\ = \tan A + \tan B - \tan A - \tan B \\ = 0$$

$\ddot{x} = 0$
 $\dot{x} = c_1$
 When $t=0$, $\dot{x} = V \cos \alpha$
 $\therefore c_1 = V \cos \alpha$
 $\therefore \dot{x} = V \cos \alpha$
 $x = Vt \cos \alpha + c_2$
 When $t=0$, $x=0$
 $\therefore c_2 = 0$
 $\therefore x = Vt \cos \alpha$

$\ddot{y} = -10$
 $\dot{y} = c_3 - 10t$
 When $t=0$, $\dot{y} = V \sin \alpha$
 $\therefore c_3 = V \sin \alpha$
 $\therefore \dot{y} = V \sin \alpha - 10t$
 $y = Vt \sin \alpha - 5t^2 + c_4$
 When $t=0$, $y=0$
 $\therefore c_4 = 0$
 $\therefore y = Vt \sin \alpha - 5t^2$

(ii) When $x=p$, $y=h$

$$\therefore t = \frac{p}{V \cos \alpha}$$

$$\text{and } h = V \sin \alpha \cdot \frac{p}{V \cos \alpha} - 5 \cdot \frac{p^2}{V^2 \cos^2 \alpha}$$

$$\frac{5p^2}{V^2 \cos^2 \alpha} = p \tan \alpha - h$$

$$\frac{V^2 \cos^2 \alpha}{5p^2} = \frac{1}{p \tan \alpha - h}$$

$$V^2 = \frac{5p^2 \sec^2 \alpha}{p \tan \alpha - h}$$

$$= \frac{5p^2(1 + \tan^2 \alpha)}{p \tan \alpha - h}$$

(iii) Similarly $V^2 = \frac{5q^2(1 + \tan^2 \alpha)}{q \tan \alpha - h}$

$$\therefore \frac{5p^2(1 + \tan^2 \alpha)}{p \tan \alpha - h} = \frac{5q^2(1 + \tan^2 \alpha)}{q \tan \alpha - h}$$

$$p^2(q \tan \alpha - h) = q^2(p \tan \alpha - h)$$

$$(p^2 q - q^2 p) \tan \alpha = (p^2 - q^2) h$$

$$\tan \alpha = \frac{(p+q)(p-q)h}{pq(p-q)}$$

$$= \frac{h(p+q)}{pq}$$